

MS:

12.1

$$(a) \begin{pmatrix} 10 \\ 2 \\ 5 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 15 \\ 6 \\ 8 \end{pmatrix} = \begin{pmatrix} 14 \\ 17 \\ 11 \\ 10 \end{pmatrix}$$

(b) Not possible.

$$(c) \begin{pmatrix} -10 \\ -2 \\ -5 \\ -2 \end{pmatrix}$$

$$(e) -3f = \begin{pmatrix} -60 \\ -12 \\ -30 \\ -12 \end{pmatrix}$$

$$(f) \|b\| = \sqrt{4^2 + 15^2 + 6^2 + 8^2} = \sqrt{341}$$

12.3

$$(a) A' = \begin{pmatrix} 0 & 1 & 5 \\ 1 & -2 & -1 \\ 5 & -1 & 2 \end{pmatrix}$$

$$(b) B' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(c) C' = \begin{pmatrix} 1 & 3 \\ 1 & -2 \end{pmatrix}$$

$$(d) D' = \begin{pmatrix} 0 & 5 & 2 & 1 \\ 1 & 1 & 4 & 1 \\ 2 & -1 & 0 & 0 \end{pmatrix}$$

12.4

(a) Not conformable to addition

$$(b) A - B = \begin{pmatrix} 2 & -3 & -3 \\ 8 & 5 & -3 \end{pmatrix}$$

$$(c) A + 5B = \begin{pmatrix} 20 & 21 & 27 \\ -4 & -13 & 33 \end{pmatrix}$$

$$(d) 3A = \begin{pmatrix} 15 & 3 & 6 \\ 18 & 6 & 9 \end{pmatrix}$$

$$(e) 2B - 5A = \begin{pmatrix} -19 & 3 & 0 \\ -34 & -16 & -3 \end{pmatrix}$$

$$(f) \begin{pmatrix} 3 & -2 \\ 4 & -3 \\ 5 & 6 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ -5 & 3 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -4 \\ 9 & -6 \\ 8 & 5 \end{pmatrix}$$

$$12.5 \quad (a) |A| = 2 - (-4) = 6$$

$$(b) |B| = 12 - 12 = 0$$

(c) Cannot take the determinant of a non-square matrix

$$(d) |D| = 3 \begin{vmatrix} -5 & 1 \\ 2 & 3 \end{vmatrix} - 2 \begin{vmatrix} -1 & 1 \\ 3 & 3 \end{vmatrix} + (-4) \begin{vmatrix} -1 & -5 \\ 3 & 2 \end{vmatrix} \\ = 3(-15 - 2) - 2(-3 - 3) - 4(-2 + 15) = 3(-17) - 2(-6) - 4(13) = -91$$

$$(e) |E| = \begin{vmatrix} 3 & 2 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{vmatrix} = 3(0 - 1) - 2(0) + 0 = -3$$

$$12.6 \quad (a) |A| = 0 \Rightarrow \text{Matrix is singular, no inverse}$$

$$(b) |B| = 2 - 12 = -10$$

$$\text{Finding cofactors: } C = \begin{vmatrix} 2 & -3 \\ -4 & 1 \end{vmatrix}$$

$$\text{Adjoin B} = C' = \begin{vmatrix} 2 & -4 \\ -3 & 1 \end{vmatrix} \Rightarrow B^{-1} = \frac{\text{Adjoin B}}{|B|} = \frac{-1}{-10} \begin{vmatrix} 2 & -4 \\ -3 & 1 \end{vmatrix}$$

13.1

$$(a) sA + tB = \begin{pmatrix} 10s + 4t \\ 2s + 15t \\ 5s + 6t \\ 2s + 8t \end{pmatrix}$$

(b) At least two: $e = a+b$, $f=2a$

$$13.2 \quad (a) |A| = \begin{vmatrix} 1 & -3 & 6 \\ 2 & -5 & 10 \\ 3 & -8 & 17 \end{vmatrix} \Rightarrow |A| = -5 + 12 - 6 = 1$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

$$C = \begin{bmatrix} -5 & -4 & -1 \\ 3 & -1 & -1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$C' = \text{Adjoint}A = \begin{bmatrix} -5 & 3 & 0 \\ -4 & -1 & 2 \\ -1 & -1 & 1 \end{bmatrix}$$

$$X^* = \frac{\text{Adjoint}A}{|A|} \cdot d = \begin{bmatrix} -5 & 3 & 0 \\ -4 & -1 & 2 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

(b) Using elementary row operations.

$$\left[\begin{array}{ccc|ccc} 4 & 2 & -3 & 1 & 0 & 0 \\ 6 & 3 & -5 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow R_1 = -R_1 + 2R_3 \text{ and } R_2 = R_2 - 6R_3$$

$$\left[\begin{array}{ccc|ccc} -2 & 0 & 7 & -1 & 0 & 2 \\ 0 & -3 & -17 & 0 & 1 & -6 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow R_3 = \frac{1}{2}R_1 + R_3$$

$$\left[\begin{array}{ccc|ccc} -2 & 0 & 7 & -1 & 0 & 2 \\ 0 & -3 & -17 & 0 & 1 & -6 \\ 0 & 1 & \frac{11}{2} & -\frac{1}{2} & 0 & 2 \end{array} \right]$$

$$\rightarrow R_3 = \frac{1}{3}R_2 + R_3$$

$$\left[\begin{array}{ccc|ccc} -2 & 0 & 7 & -1 & 0 & 2 \\ 0 & -3 & -17 & 0 & 1 & -6 \\ 0 & 0 & -\frac{1}{6} & -\frac{1}{2} & \frac{1}{3} & 0 \end{array} \right]$$

$$\rightarrow R_3 = -6R_3$$

$$\left[\begin{array}{ccc|ccc} -2 & 0 & 7 & -1 & 0 & 2 \\ 0 & -3 & -17 & 0 & 1 & -6 \\ 0 & 0 & 1 & 3 & -2 & 0 \end{array} \right]$$

$$\rightarrow R_1 = R_1 - 7R_3 \text{ and } R_2 = R_2 + 17R_3$$

$$\left[\begin{array}{ccc|ccc} -2 & 0 & 0 & -22 & 14 & 2 \\ 0 & -3 & 0 & 51 & -33 & -6 \\ 0 & 0 & 1 & 3 & -2 & 0 \end{array} \right]$$

$$\rightarrow R_1 = -\frac{1}{2}R_1 \text{ and } R_2 = -\frac{1}{3}R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 11 & -7 & -1 \\ 0 & 1 & 0 & -17 & 11 & 2 \\ 0 & 0 & 1 & 3 & -2 & 0 \end{array} \right]$$

Solve for $\mathbf{x}^* = A^{-1}d$

$$\mathbf{x}^* = \begin{bmatrix} 11 & -7 & -1 \\ -17 & 11 & 2 \\ 3 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Cont.

- (c) Two equations and three unknowns, no unique solution. Setting $z = t$: the second equation gives $y = 2x + t - 1$; substituting into the first gives $8x + 10t = 10$, so

$$(x, y, z) = \left(\frac{5}{4} - \frac{5}{4}t, \frac{3}{2} - \frac{3}{2}t, t\right), \quad t \in \mathbb{R}.$$

- 13.4 (a) $|D| = -2 \neq 0 \Rightarrow D$ is full rank

- (b) $|E| = 0 \Rightarrow$ rows are linearly dependent $\Rightarrow$ rank is 1

13.5 (a) $|A| = \begin{vmatrix} 1 & -1 & 2 \\ -1 & 2 & 3 \\ 1 & 1 & -1 \end{vmatrix} = -13$

$$|A_1| = \begin{vmatrix} 4 & -1 & 2 \\ 1 & 2 & 3 \\ 5 & 1 & -1 \end{vmatrix} = -54$$

$$|A_2| = \begin{vmatrix} 1 & 4 & 2 \\ -1 & 1 & 3 \\ 1 & 5 & -1 \end{vmatrix} = -20$$

$$|A_3| = \begin{vmatrix} 1 & -1 & 4 \\ -1 & 2 & 1 \\ 1 & 1 & 5 \end{vmatrix} = -9$$

$$x = \frac{|A_1|}{|A|} = \frac{-54}{-13} = \frac{54}{13}, \quad y = \frac{20}{13}, \quad z = \frac{9}{13}$$

(b) $|A| = \begin{vmatrix} 3 & -2 & -1 \\ -3 & 2 & -5 \\ 2 & 2 & -2 \end{vmatrix} = 60$

$$|A_1| = \begin{vmatrix} 4 & -2 & -1 \\ 2 & 2 & -5 \\ 5 & 2 & -2 \end{vmatrix} = 72$$

$$|A_2| = \begin{vmatrix} 3 & 4 & -1 \\ -3 & 2 & -5 \\ 2 & 5 & -2 \end{vmatrix} = 18$$

$$|A_3| = \begin{vmatrix} 3 & -2 & 4 \\ -3 & 2 & 2 \\ 2 & 2 & 5 \end{vmatrix} = -60$$

$$x = \frac{72}{60} = \frac{6}{5}, \quad y = \frac{18}{60} = \frac{3}{10}, \quad z = \frac{-60}{60} = -1$$